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NMAM INSTITUTE OF TECHNOLOGY, NITTE*Off-Campus Centre of Nitte (Deemed to be University)***First / Second Semester B.Tech (CBCS) Degree Examinations**

Summer Semester – July 2024

ME1003-2 – ELEMENTS OF MECHANICAL ENGINEERING

(ACT,BT,CV,EC,EE,ME,VLSI,)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **The application of artificial intelligence to analyse sensor data and detect failure patterns in machinery and components is called as**
A) Predictive maintenance
B) IIOT
C) 5G and edge computing
D) Data analytics
2. **Which of the following is not a bio-material**
A) Titanium
B) Silver
C) Gold
D) Cast iron
3. **The minimum wind speed required generate electricity in a wind mill is**
A) 8 to 10 km/h
B) 15 to 18 km/h
C) 12 to 14 km/h
D) Above 22 km/h
4. **The ability of the material to absorb the energy is called**
A) Resilience
B) Creep
C) Toughness
D) Stiffness
5. **Francis turbine is a**
A) Low head low discharge
B) High head low discharge
C) High head high discharge
D) Medium head high discharge
6. **Centrifugal pump works on the principle of**
A) Centrifugal force
B) Second law of thermodynamics
C) Forced vortex flow
D) None of the above
7. **In Parson's steam turbine, the primary driving force is**
A) Sum of reaction and impulsive force
B) Impulse force
C) Reaction force
D) Centrifugal force
8. **In reaction steam turbine, pressure**
A) decreases over all the blades
B) decreases over moving blades
C) decreases over fixed blades
D) increases over all the blades
9. **In 4 stroke I.C engine, each stroke is completed when crankshaft rotates by**
A) 90°
B) 180°
C) 270°
D) 360°
10. **Which of the following is not an ideal property of a refrigerant**
A) It must have low boiling point
B) It must have a very low freezing point
C) It must have a very low thermal conductivity
D) It must have a low specific volume

11. Ration of compression ratio for petrol engine ranges from
 A) 1:7 to 1:11 B) 1:3 to 1:10
 C) 1:8 to 1:15 D) 1:15 to 1:20
12. In an I C engine, the load on brake drum is 35kg, radius of the brake drum is 0.6m, then the torque on the brake drum is
 A) 0.303kNm B) 0.256kNm
 C) 303kNm D) 0.206kNm
13. ----- gears are used to transmit the power between two shafts having their axes at right angles and non-coplanar.
 A) Bevel B) Rack and pinion
 C) Worm and worm wheel D) helical
14. Two gears having teeth 80 and 30 mesh with each other. If the smaller gear wheel runs at 480rpm, then the speed of the larger will be
 A) 480rpm B) 180rpm
 C) 1280rpm D) 520rpm
15. Non consumable electrode is used in
 A) MIG B) TIG
 C) Arc welding D) Gas welding
16. Which of the following flame is used for cutting in gas welding
 A) Oxidizing flame B) Carburizing flame
 C) Neutral flame D) None of the above
17. A system that is capable of producing a variety of products (or parts) with virtually no time lost for changeovers from one product to the next.
 A) Flexible automation B) Programmable automation
 C) Fixed automaton D) All the above
18. Controlling and intelligence of the robots are dealt by
 A) Computer engineering B) Electrical engineering
 C) Mechanical engineering D) civil engineering
19. Which of the following operation does Not use a rotating tool?
 A) Reaming B) Counter - Boring
 C) Facing D) End Milling
20. Machining process to produce internal threads is
 A) Counter boring B) Tapping
 C) Thread cutting D) Reaming

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Describe the formation of steam at constant pressure.	5	L*1	1	1
	b) Discuss the emerging trends in automotive industry.	3	L1	1	1
	c) With neat sketch explain the working of impulse steam turbine.	8	L1	2	1
2.	a) What are bio materials? Write the application of same.	03	L1	1	1
	b) Draw the stress strain curve for mild steel and explain the salient points.	05	L1	1	2
	c) With neat sketch explain the working of Francis turbine	08	L1	2	1
3.	a) Explain				
	i) Stress				
	ii) Toughness				
	iii) convection	03	L1	1	1
	b) Explain the working of a thermal power plant with a neat sketch.	05	L1	1	1
	c) Differentiate between open cycle and closed cycle gas turbine.	03	L1	2	1

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| d) Explain the working of a single stage compressor with neat sketch. | 05 | L1 | 2 | 2 |
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Unit – II

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| 4. a) With neat sketch explain the working of 4-stroke diesel engine | 08 | L1 | 3 | 1 |
| b) Differentiate between soldering and welding. | 03 | L1 | 4 | 1 |
| c) In a crossed belt drive difference in tension between the tight and slack side of the belt is 1000N. Find the tension on tight side and slack side if angle of contact is 160° and coefficient of friction is 0.3. | 05 | L2 | 4 | 2 |
| 5. a) Determine the brake power of a 4 stroke IC engine, Given
Diameter of brake drum=1.5m, spring balance reading =5kg, diameter of rope=10mm, crankshaft speed=200rpm, load suspended on brake drum=100kg. | 08 | L2 | 3 | 1 |
| b) Differentiate between welding, soldering and Brazing. | 08 | L1 | 4 | 1 |
| 6. a) Discuss the primary components of electrical vehicle. | 04 | L1 | 3 | 1 |
| b) Differentiate between petrol and diesel engine. | 04 | L1 | 3 | 1 |
| c) a) State when the following types of gear drives are employed?
(i) Helical gears (ii) Bevel gears | 02 | L1 | 4 | 2 |
| d) Explain the process of TIG welding in detail. | 06 | L1 | 4 | 1 |

Unit – III

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| 7. a) Explain how Taper turning is performed on lathe. | 08 | L1 | 5 | 1 |
| b) Draw and explain the different robot configuration | 08 | L1 | 5 | 1 |
| 8. a) With an example explain closed loop and open loop control system | 06 | L1 | 5 | 1 |
| b) Explain the following operation
i) plane milling
ii) counter sinking | 06 | L1 | 5 | 1 |
| c) Describe the characteristics of flexible automation and Discuss some examples of situations in which its use would be beneficial. | 04 | L2 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
